A Tale of One Exchange and Two Order Books: Effects of Fragmentation in the Absence of Competition

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Abstract

Equity exchange operators nowadays commonly run multiple limit order books, thus creating a higher degree of fragmentation than that of competition for trading. The effects of market fragmentation and market competition in equity markets are difficult to disentangle and this distinction has not been made in previous literature. We isolate the role of market fragmentation both theoretically and empirically. We develop a dynamic model of multiple limit order markets, in which traders make endogenously optimal decisions to maximize their expected payoffs, taking into account intrinsic incentives, markets conditions, potential future trading decision and strategies adopted by different agents. The model contains no strategic behavior on the part of exchanges allowing us to compare market outcomes when trading is conducted on two limit order book with a consolidated market. In this sense, we observe the effects of fragmentation while holding competition constant. Empirically, we exploit a decision by Euronext to eliminate duplicate limit order books for stocks that are listed, for historical reasons, in more than one of its countries of operation. This decision reduces market fragmentation without affecting market competition. Our combination of theoretical and empirical results suggests that, holding competition between market operators constant, consolidated markets offer lower transaction costs, lower profits for intermediaries, higher market efficiency, and higher aggregate welfare than fragmented markets. These results suggest that competition in market design, not market fragmentation, drives previous findings of market quality improvements when new trading venues emerge. Our study has policy implications concerning the operation of multiple order books by individual exchange operators.

Keywords: Fragmentation, Competition, Liquidity, Price Efficiency

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1. Introduction

Fragmentation of trading activity has been one of the most significant changes in the equity market structure in recent years. Equity markets in the US, EU and elsewhere have moved from monopolist or dominant exchanges to a competitive market where multiple exchange and alternative trading systems coexist.

This competition has been enabled by regulatory changes such as Regulation National Market System (RegNMS) in the United States and the Markets in Financial Instruments Directive (MiFID) in the European Union. These changes have brought forth a wealth of research on the trade-off between innovation and cost reduction arising from the existence of competition, and potentially detrimental effects on liquidity due to network externalities. In this literature, the two concepts of fragmentation and competition have been examined. On the one hand, in order to establish superiority, trading venues offer alternative features to their customers, such as lower (sometimes negative) trading fees, higher trading speeds, and the ability to execute large blocks. On the other hand, fragmentation of trading generally refers to trading a security on multiple venues without specifically referring to the types of venues involved. These two issues are naturally intertwined as an increase in one generally leads to an increase also in the other. When a new venue starts to successfully compete with an incumbent venue, markets fragment. However, as the number of competing market operators has increased, equity markets have simultaneously experienced a process of consolidation as a result of national and international mergers of exchange operators.

This has led to a situation in which, while there exist a host of non-exchange trading venues characterized by a lack of pre-trade transparency, only a small number of exchange operators compete with each other while operating several exchanges each. For example, in the United States, the three large exchange operators - Intercontinental Exchange, Nasdaq OMX, and BATS - currently operate a total of ten lit equity exchanges. Consequently, equity markets experience a divergence in the degrees of competition and fragmentation. This raises the question as to possible effects of fragmentation when competition remains unaffected.[1]

In this paper, we address this question by studying the period around NYSE Euronext's implementation of a single order book on 14 January 2009 for their Paris, Amsterdam, and

[1] It is possible that exchange operators allow a certain degree of competition between the different exchanges they own, although it appears implausible that such competition would be similar to that between exchanges run by different owners.

Brussels markets. Before this date, the exchange operated multiple independent order books for stocks which were cross-listed on these markets. As these order books had the exact same trading protocols, the introduction of a single order book led to a decrease in the fragmentation of the stocks without any corresponding change in the competitive environment.[2]

We evaluate this shock to the trading activity of the affected stocks. There are two main advantages of examining this event. First, as it was based on a business decision by Euronext, there was no possibility of any selection bias, as all multi-listed stocks experienced the same treatment. Second, the announcement of this event was made more than one year before the event date in order to allow market participants to adapt and test their trading systems. This eliminates potential concerns about the event date confounding with other market events around the same time.[3] We exploit this event to study the effects of fragmentation on market quality in modern electronic limit order markets, abstracting away any changes to competition between markets.

Goettler et al. (2005, 2009) model a dynamic limit order market and examine the welfare implications of specific design choices such as minimum tick sizes and asymmetric information, respectively. We extend their model to a case where a single asset trades in a fragmented market with two limit order books. In our model traders arrive in the market and make endogenously optimal trading decisions to maximize expected payoffs, taking into account intrinsic incentives, markets conditions, potential future trading decision and strategies adopted by different agents. While the modeling framework is flexible enough to allow for alternative market design choices, we study the impact in a case where all design parameters such as trading fees, tick sizes, and transparency are constant in both limit order books. This is similar to the secondary market setting for stocks with listings in multiple Euronext markets. We compare the fragmented market to a case where the same security trades in a single consolidated market and derive testable predictions on the likely impact of consolidation on trader behaviour, liquidity, price efficiency, and welfare.

The model generates three main testable predictions. First, the model predicts that overall liquidity is lower in fragmented markets. Specifically, quoted and effective spreads in each local market are higher relative to the consolidated market case. This is because, in a consolidated

[2]Cross-listed firms had to choose one Market-of-Reference (MoR) which continued operating after the implementation of a single order book.

[3]Although the original date of implementation was postponed, this was mainly due to technical reasons as opposed to concerns about market conditions. In any case, the final implementation date was announced more than 60 days in advance.

market, all the order flow is routed to the single order book. In order to improve their probability of execution liquidity providers have to place more aggressive orders. This generates higher price competition translating into improved spreads. The model predicts market depth to be affected in two opposite directions. On the one hand, quoted depth in each local market is reduced as execution probabilities are lower due to market orders being routed to multiple markets. This leads to less limit orders being routed to each local market. On the other hand, consistent with the existing literature such as Foucault and Menkveld (2008) and Degryse et al. (2015), consolidated depth is higher in fragmented markets.

Second, the model also predicts a decrease in price efficiency in fragmented markets. This is because small price dislocations relative to the fundamental value are more likely to occur when order flow is split across multiple venues. This prediction is consistent with Madhavan (1995) who predicts higher price volatility in fragmented markets.

Finally, market fragmentation leads to a slight decrease in overall welfare. More importantly, the distribution of welfare gains between agents who trade for intrinsic motives (predominantly consumers of liquidity) and those who do not (liquidity providers) change significantly. While the overall trading strategies employed by these agents do not change, liquidity providers manage to extract much higher gains from trading in fragmented markets. This is because liquidity providers can circumvent time priority in one order book by submitting limit orders to the second order book. This allows them, in equilibrium, to post less aggressive orders relative to the case involving consolidated markets. Liquidity demanders, who trade-off demand for immediacy with waiting costs, are willing to incur higher immediacy costs in order to extract the gains from trade.

We test the above three hypotheses around NYSE Euronext's introduction of a single order book. We construct a matched control group based on the stock price and market capitalization, similar to Davies and Kim (2009), in order to control for overall market conditions as well as other market-wide changes implemented by NYSE Euronext. Our sample consists of nine stocks each in the treatment and control group and our time period covers all trading days between 1 December 2008 and 28 February 2009. We compare the impact of the introduction of single order book on trading volume, liquidity, and price efficiency. We measure liquidity by computing quoted spread, top-of-book depth, effective spread, price impact, and realized spread, and we measure price efficiency by computing autocorrelations and variance ratios, for stocks in the treatment and control group.

Before the introduction of a single order book, 40% of the total Euronext trading was generated in the less active order book. Overall, the elimination of this order book represents a very large shock to the level of fragmentation as compared to other important events affecting market fragmentation. For example, Foucault and Menkveld (2008) observe that Euronext lost approximately 3.5% market share after EuroSETS' entry into the Dutch stocks. Hengelbrock and Theissen (2009) observe that Turquoise gained between 0% and 7% market share after entry into the different European markets.

Consistent with the theoretical predictions, after introduction of a single order book, we find that quoted liquidity improves compared to that earlier observed in the local markets. The most active of the two order books experiences a 27% decrease in quoted spreads, whereas top-of-book depth increases by almost 50% for the largest stocks. We also find that, as opposed to the theoretical predictions, consolidated depth is higher in the single market although this effect is only significant for the most liquid stocks in the sample. Local effective spreads also declined by 37% after the event. However, effective spreads measured relative to the inside mid-quotes do not experience any change. Realized spreads - which measure liquidity providers' compensation after eliminating adverse selection costs - decreased, depending on the specification, between 38% and 73%. This strong result is also consistent with our theoretical model which predicts that market-makers earn more profits in fragmented markets. We also observe that price impacts relative to inside quotes increase after the event, although they are statistically significant only for the most liquid stocks.

We also find that short-term price efficiency improves although the improvements are generally statistically insignificant. All combinations of autocorrelations and variance ratios get closer to one, which is consistent with prices becoming closer to resembling a random walk.

While we are unable to empirically compute welfare in the absence of information on traders' private values, we find that the introduction of a single order book leads to a weakly significant increase in trading volume. This is despite the elimination of arbitrage trades, which amount to 7.8% of the total trading volume during the pre-event period. This points to an increase in overall welfare due to reduced intermediation in the consolidated market.

While theoretical models separately examine the effects of fragmentation and competition, the empirical literature necessarily focuses on the combined effects of the two. The theoretical literature suggests that liquidity should concentrate in one central market unless different markets have distinct characteristics, for example, with respect to transparency and anonymity, making them attractive to different groups of traders. Mendelson (1987) emphasizes the negative effects of fragmentation in specialist markets and predicts lower trading volume, increased price volatility, and lower welfare because of network externalities. Pagano (1989) suggests that an equilibrium with multiple markets is possible only if trading costs and absorptive capacity differ, allowing a separate market for large traders to coexist. Chowdhry and Nanda (1991) show that in the presence of different transaction costs liquidity traders and informed traders choose the market with least transaction costs, resulting in concentration. Biais (1993) compares centralized limit-order or specialist markets with decentralized opaque dealer markets and finds the number of dealers to be increasing in trade frequency and price volatility and spreads to be equal in expectation but more volatile in the centralized market.

Madhavan (1995) argues that markets fragment only if there is a lack of trade disclosure. Fragmentation in his model benefits dealers and large traders, and increases volatility and price inefficiency. Parlour and Seppi (2003) study competition between a limit-order market and a hybrid limit-order/specialist market and show that either market can be dominant, though coexistence is also possible. In possibly the most relevant study to today's competitive landscape of equity markets, Foucault and Menkveld (2008) model competition between two limit order books and predict that the entry of a second market increases consolidated depth, and that increased use of smart order routers leads to an increase in liquidity in the entrant market. They also test their model empirically on a sample of Dutch stocks and find that the market entry of the London Stock Exchange increased global depth.

Three empirical studies closest to our paper are Amihud et al. (2003), Boehmer and Boehmer (2003) and Nguyen et al. (2007). Amihud et al. (2003) study the reduction in fragmentation on the Tel Aviv Stock Exchange upon exercise of deep in-the-money warrants at maturity and find that stock prices increase and stock liquidity improves. Boehmer and Boehmer (2003) and Nguyen et al. (2007) examine the impact of NYSE's entry in the ETF market and find improvements in different measures of liquidity.

In other studies, Hengelbrock and Theissen (2009) and Chlistalla and Lutat (2011) analyze the market entries of Turquoise and Chi-X, respectively, and find that this event positively affected liquidity in the main market. Riordan et al. (2010) find that MTFs contribute to the majority of quote-based price discovery for the FTSE100 stocks in the UK. Kohler and von Wyss (2012) and Hellström et al. (2013) find that fragmentation in the Swedish market increases liquidity, for all but large stocks, and price efficiency for all stocks respectively. O'Hara and Ye

(2011) analyze overall fragmentation in the US equity markets and find that it is not harmful to market quality. Degryse et al. (2015) and Gresse (2014) differentiate between fragmentation in lit and dark markets and find that lit fragmentation improves liquidity, while disagreeing on the effects of dark fragmentation.[4]

Our results suggest that the loss of positive network externalities as a result of fragmentation are substantial even in today's modern limit order markets. This is in spite of the presence of high-frequency traders that serve to integrate disparate equity markets through their arbitrage activities. Our results also have policy implications. Market operators nowadays tend to run multiple exchanges in European and US equity markets. The functionality of these exchanges in some cases does not fundamentally differ with the exception of pricing schedules. This leaves markets more fragmented than would be necessary for a given level of competition. One may conclude that preventing market operators from operating multiple markets without a meaningful differentiation between their services would benefit investors.

2. Model

We develop a dynamic continuous-time model of a single financial asset that trades in one or two pure limit order markets. In the model, agents with heterogeneous private values make endogenously sequential optimal decisions in continuous time, to maximize their expected payoffs, taking into account markets conditions, potential future trading decisions, and different strategies adopted by other agents. As in real limit order markets, the limit orders books are characterized by a set of discretized prices at which traders can submit orders that respect time and price priority for execution of limit orders. In this section, we provide a description of the model.

2.1 Model Components

Limit Order Books: Each market $m \in \{1, 2\}$ has a respective order book. As in real limit order markets, the limit order book relative to the market m at time t, $L_{m,t}$ is described by a discrete set of feasible prices, denoted as $\{p_m^i\}_{i=-\infty}^{\infty}$, where the tick size, d_m is the distance between any two consecutive prices. There are backlogs of outstanding orders to buy or to sell in the market m, $l_{m,t}^i$, at prices p_m^i. A positive (negative) number in $l_{m,t}^i$ denotes buy (sell) orders, and it represents the depth at price p_m^i. Therefore, for the book $L_{m,t}$, the bid price is

[4]See Gomber et al. (2016) for a more extensive review of the literature on the competition and fragmentation of equity markets.

$B(L_{m,t}) = \sup\{p_j^i | l_{m,t}^i > 0\}$, while the ask price is $A(L_{m,t}) = \inf\{p_j^i | l_{m,t}^i < 0\}$, and if the order book is empty on the buy side or on the sell side, $B(L_{m,t}) = -\infty$ or $A(L_{m,t}) = \infty$.[5]

Each limit order book respects price and time priority while executing limit orders. Buy (sell) limit orders at higher (lower) prices are executed first, and limit orders submitted earlier have priority in the queue when they have the same price. In addition, the limit price identifies whether an order is marketable or not. This means that an order to buy (sell) at a price above (below) the best ask (bid) price is marketable and executed immediately at the ask (bid) price(s) in the queue.

Asset: The financial asset has a fundamental value v_t at the time t, which can be interpreted as the present value of future cash flows, and follows a Poisson process at rate λ_v. Whenever a change occurs, the fundamental value increases or decreases by a fixed amount f, each with the same probability. Differently to Goettler et al. (2009), when a change occurs we do not assume that the fundamental value coincides with a feasible price on the pricing grid, therefore the fixed amount f does not depend on the tick sizes d_m. This is important since each market could have different tick sizes (and hence different pricing grids) but v_t can take any value according to the Poisson process described above.

Agents: Risk-neutral agents arrive at the market randomly following a Poisson process with intensity λ. Each agent has a private value α for the asset, which is drawn from a discrete distribution F_α and known *ex ante*. This private value can be interpreted as a personal valuation of the asset and is constant for each agent.[6] These private values introduce heterogeneity across different agents in the dynamic trading game. For instance, traders with $\alpha > 0$ have a positive valuation for the asset and therefore they are more likely to be buyers. Analogously, traders with $\alpha < 0$ are more likely to be sellers. Traders with zero private value have no intrinsic benefits to trade and maximize their benefits depending on the available trading possibilities. Hence they are indifferent with respect to taking either side of the market and more likely to act as liquidity suppliers by submitting limit orders.[7] Conversely, traders with higher absolute private values are more likely to demand liquidity by submitting market orders.

Each agent can trade one share,[8] and has to select the best action given the current state

[5] We use a similar notation to Goettler et al. (2009) regarding the microstructure features of the model for the dynamic order book market.

[6] Private value represents idiosyncratic motives to trade such as wealth shocks, hedging needs, tax exposures, differences in investment horizons, among others.

[7] Jovanovic and Menkveld (2011) also define liquidity providers as agents with zero private values.

[8] We can include additional shares per agent in the trading decision. However, similarly to Goettler et al. (2009), we assume one share per trader to make the model computationally tractable.

of the economy. To this end a trader has to make four main trading decisions after arriving at the market: i.) to submit an order or to wait until the market conditions change; ii.) to trade in $L_{1,t}$ or $L_{2,t}$; iii.) to buy or to sell the asset; iv.) to choose the price at which he will submit the order, which implies the decision to submit a market order or a limit order. Despite the fact that traders arrive following a Poisson process, the submission rate is different as agents can decide whether to submit an order, which depends endogenously on the market conditions at the time t.

Agents can re-enter the market and monitor their previously unexecuted limit orders. However, they cannot immediately modify their existing limit orders after a change in the market conditions, mainly due to cognition limits.[9] Traders re-enter the market according to a Poisson processes with rate λ_r and have to make the following additional trading decisions after re-entering: i.) whether to cancel an unexecuted limit order or retain the order without changes; ii.) if he decides to cancel the order, whether to submit a new order for the asset or wait for different market conditions in the future; iii.) if he decides to submit a new order after a cancellation, he has to choose which market to send the new order, the type of order (buy or sell) and its price. Thus, agents have to take the possibility of re-entry into account in their utility maximization problem. Once a trader submits a limit order, he remains part of the trading game until it is executed; however, after execution the trader exits the market forever. Consequently, there are a random number of active market participants at each instant who are monitoring their previous limit orders.

Traders have to pay a cancellation fee c_m when they cancel an existing limit order in $L_{m,t}$, $m \in \{1, 2\}$. In the case of a re-entry, a trader can leave the order unchanged, which has the benefit of keeping her priority time in the respective queue and avoiding a cancellation fee.[10] The costs associated with leaving an order in the book are related to changes in the asset value affecting future payoffs. For instance, in case of an increase in asset value, some stale limit sell orders priced too low could be 'picked off' by a fast trader in order to profit from the difference. Conversely, when the asset value decreases, a limit sell order priced too high has the risk of non-execution. To take into account the risk that a limit order may not result in a trade, we include a cost of 'delaying' by a discount rate ρ, which is common for both markets and reflects

[9] Roşu (2009) proposes a dynamic model of a limit order book in which traders can modify instantaneously their previous orders.

[10] It is important to point out that the order priority can still change, depending on the shape of the book, which will be taken into account in the decision to cancel and re-submit.

the cost of not executing immediately. This does not represent the time value of money; instead ρ reflects opportunity costs and the cost of monitoring the market until a limit order is executed. Thus, the payoffs of order executions are discounted back to the order submission time at rate ρ.

2.2 Agents' Dynamic Maximization Problem

Suppose that a trader arrives in the market at time t and observes state s of the market. The state s that a given trader observes includes the state of the two limit order books $L_{1,t}$ and $L_{2,t}$ (quotes, market depths, last transaction) that result from previous trading activity in each market, the current fundamental value of the asset v_t, her private value α and delay rate ρ, the status of her previous action in case she submitted an earlier limit order to any of the markets including the original submission price, the book where the order was submitted, the current priority in the book, and the direction.

Define $\Gamma(s)$ as the set of possible of actions that a trader can take given the state s (e.g. to send an order to the limit order book $L_{1,t}$ or $L_{2,t}$, to wait until market conditions change, to cancel and submit a new order, among others). Formally, we define an action as $\tilde{a} = (\tilde{b}, \tilde{p}, \tilde{x}, \tilde{q})$ where $\tilde{b}$ represents the optimal limit order book chosen, $\tilde{p}$ is the price of the order, $\tilde{x}$ is $+1$ or -1 when the order is a buy or sell, and 0 otherwise, and $\tilde{q} \geq 0$ the priority associated to this order, which is determined by $\tilde{p}, \tilde{x}$ and $L_{\tilde{b},t}$. Let $\eta(t|\tilde{a}, s)$ be the probability that an order is executed at time t given that the trader takes action $\tilde{a} \in \Gamma(s)$ while facing state s. It is important to note that $\eta(\cdot)$ incorporates all possible future states and strategic actions adopted by other traders until t. If the decision $\tilde{a}$ is the submission of a market order, $\eta(0|\tilde{a}, s) = 1$, while $\eta(h|\tilde{a}, s)$ converges asymptotically to zero when the trader decides to submit a limit order with a price far away from the fundamental value (not an aggressive order). In addition, let $\gamma(v|t, s)$ be the density function of v at time t and state s. Therefore, the expected value of an order that is executed prior to agent's re-entry at time t_r is:

$$\pi(t_r, \tilde{a}, s) = \int_0^{t_r} \int_{-\infty}^{\infty} e^{-\rho t} \left((\alpha + v - \tilde{p})\tilde{x}\right) \cdot \eta(t|\tilde{a}, s) \cdot \gamma(v|t, s) dv dt \tag{1}$$

Here, $(\alpha + v - \tilde{p})\tilde{x}$ is the instantaneous payoff of the order where $\tilde{p}$ is the submission price which is part of the decision $\tilde{a}$; while $\tilde{x}$ is also a component of the decision $\tilde{a}$ and reflects whether the trader decides to submit a buy order ($\tilde{x} = 1$), to submit a sell order ($\tilde{x} = -1$) or to submit no order ($\tilde{x} = 0$). This payoff is transformed to a present value at the rate ρ which is the cost of delaying defined previously.

Let $R(\cdot)$ be the probability distribution of the re-entry time which is exogenous and follows an exponential distribution at rate λ_r. In addition, let $\psi(s_{t_r}|\tilde{a}, s, t_r)$ be the probability that the state s_{t_r} takes place at time t_r given the previous state s and the action $\tilde{a}$, which includes all potential states and strategic decisions followed by other traders until t_r. Therefore, the value to an agent of arriving at the state s, $V(s)$, is given by the Bellman equation of the trader's optimization problem:

$$V(s) = \max_{\tilde{a}\in\Gamma(s)} \int_0^{\infty} \left[\pi(t_r, \tilde{a}, s) + e^{-\rho t_r} \int_{s_{t_r}\in\mathcal{S}} (V(s_{t_r}) - \tilde{z}_{s_{t_r}} c_m) \cdot \psi(s_{t_r}|\tilde{a}, s, t_r) ds_{t_r} \right] dR(t_r) \quad (2)$$

where $\tilde{z}_{s_{t_r}} = 1$ if the optimal decision in state s_{t_r} is a cancellation and $\tilde{z}_{s_{t_r}} = 0$ in any other case, $m \in \{1, 2\}$ is an indicator for the market selected previously (i.e., $L_{1,t}$ or $L_{2,t}$, respectively), and $\mathcal{S}$ is the set of possible states in re-entries. The first term is defined in equation (1); while the second term reflects the subsequent payoff in the case of re-entries.

2.3 Equilibrium

Since the model incorporates so many realistic features of limit order markets, the equilibrium does not admit an analytic and closed form expression. For that reason, we obtain the equilibrium numerically using the algorithm introduced by Pakes and McGuire (2001), originally proposed for industrial organization problems with sequential decisions. This algorithm provides a Markov-perfect equilibrium which has been successfully implanted into dynamics models for limit order markets by Goettler et al. (2009), Goettler et al. (2005) and Bernales and Daoud (2014), although without considering the possibility to trade in multi markets. We provide more details about the model implementation for two markets in Appendix A.

3. Theoretical Results

3.1 Trading Behaviour

The optimal strategies of traders are complex, because they take into account large amounts of information. On the one hand, traders can submit market orders and execute immediately, as these do not involve waiting costs. However, traders usually have to pay an immediacy cost for such trades, which is given by the difference between the fundamental value and the transaction price. On the other hand, traders can submit limit orders, which have associated waiting costs equal to the delay rate ρ. Limit orders allow agents to trade at better terms, but there is also an inherent risk of being picked off if the fundamental value changes. Limit order traders can post less aggressive quotes to reduce this picking off risk.

Observation 1. *Agents with intrinsic motives to trade act as liquidity consumers. Agents with no intrinsic motives to trade act as liquidity suppliers.*

Supplying liquidity to the markets requires posting limit orders at competitive prices. We report the average order submission strategy adopted by each trader type in Table 1. Among all limit orders in a consolidated market, 67.9% are submitted by traders with $\alpha = 0$, 26.1% by agents with $|\alpha| = 4$, and 6.1% were submitted by traders with $|\alpha| = 8$. These results are similar even in fragmented markets suggesting that fragmentation does not change the overall strategies adopted by traders. Agents with no intrinsic motive to trade submit most of the limit orders for two reasons. First, such agents speculate in both markets submitting and sometimes cancelling and re-submitting new limit orders as markets conditions change. Second, these traders are aware that other agents (with intrinsic motives to trade) are willing to pay the cost of a quick trade by demanding liquidity, which incentivizes them to be liquidity suppliers. As Table 1 reports, most market orders are submitted by agents with $|\alpha| = 8$. These results are consistent with Goettler et al. (2009) who analyse liquidity supply and demand for a single market.

3.2 Trading Gains

Consider a trader with a private value α and delaying discount rate ρ who arrives to the market at time t and executes at time t' as a consequence of her submitting a market order or through the execution of her previous limit order. Then the gross payoff or profit for the trader is calculated as:

$$\Pi = \tilde{x} \cdot (\alpha + v_{t'} - \tilde{p}) \cdot e^{-\rho(t'-t)} \tag{3}$$

where $\tilde{x} = 1$ if the trader buys and $\tilde{x} = -1$ if the trader sells, $v_{t'}$ is the fundamental value of the asset at the time of execution t' and $\tilde{p}$ is the transaction price. We define overall welfare as the mean of Π for all traders who execute.

In order to understand different sources of trading profits, we decompose the agents' payoffs to analyze gains and losses from the trading process, similar to Bernales and Daoud (2014). Suppose a trader with private value α arrives at time t and execute at time t'. Define $\Delta t = t' - t$. As we describe previously his discounted realized payoff is given by $\Pi = x \cdot ((\alpha + v_{t'} - p) \cdot e^{-\rho\Delta t}$. Suppose that $x = 1$, i.e. the order executed was a limit or market buy. Hence, we can rewrite

Equation (3) as:

$$\Pi = \alpha + \alpha(e^{-\rho\Delta t} - 1) + (v_{t'} - p)e^{-\rho\Delta t} \tag{4}$$

where we define the three terms on the right side as private value, waiting costs, and money transfer. We can consider the sell side, i.e. $x = -1$ and rewrite the discounted realized payoff for a sell order analogously. In general, a trader cannot execute immediately to gain his intrinsic private value α. Instead, traders prefer to wait, either due to lack of liquidity, poor market conditions, or the optimal action is to submit a limit order or even not an order. This is reflected in the waiting cost component.[11] Additionally, when an execution does occur, the trader further experiences gains or loses due to the difference between transaction price p and fundamental value at the moment of the trade $v_{t'}$. This is reflected in the money transfer component.

Observation 2. *(i) Global welfare of the economy decreases in the presence of market fragmentation, as opposed to a consolidated market. Agents with no intrinsic motives to trade take advantages in fragmented markets and consequently increase their profits. (ii) In fragmented markets, agents with intrinsic motives to trade experience a decrease in the absolute value of their waiting costs in exchange for worse terms on money transfer. On the other hand, agents with no intrinsic motives to trade increase their money transfer in fragmented markets, which entails a higher payoff.*

Table 2 presents the results. We report the average payoffs for each trader type in both market scenarios.[12,13] The overall result is a slight reduction in global welfare from 3.742 ticks in the single market to 3.721 in fragmented markets. This is explained mainly by the transfer of gains among the different trader types for each scenario. Agents with no intrinsic motives for trade (i.e. $\alpha = 0$) benefit in fragmented markets as they profit directly from trading activity, exploiting opportunities in both markets depending on market conditions. On the other hand, agents with intrinsic motives experience a reduction in overall welfare in a fragmented market.

We also report the decomposition of total payoff into the three components described in Equation (4). Agents with intrinsic motives to trade (i.e. $|\alpha| > 0$) exhibit a reduction in absolute waiting costs. For instance, agents with private value $|\alpha| = 4$ have a waiting cost of

[11] This interpretation for the waiting cost is similar to Hollifield et al. (2006).

[12] Note that in Table 2, the total money transfers do not add up to zero because they are discounted back to time t and Δt is different for the trader who submits the market order and the trader who submits the corresponding limit order due to traders' asynchronous arrivals. However, the instantaneous money transfer not discounted back does add up to zero.

[13] We do not report standard errors as due to the large number of new arrivals (20 million), the standard errors on the sample means are sufficiently low such that the differences in means to the order of 10^{-2} or higher are significantly different from zero.

-0.359 ticks in a consolidated market and -0.338 in fragmented markets. The corresponding values for agents with private value $|\alpha| = 8$ are -0.173 ticks and -0.127 respectively. These results suggest that agents with $|\alpha| > 0$ experience less time to execution, but as a consequence obtain worse terms of trade, which is reflected in high money transfer costs. For example, agents with private value $|\alpha| = 8$ experience a decrease in money transfer from -0.591 ticks in consolidated markets to -0.889 in fragmented markets. This is because the reduction in waiting costs does not compensate the losses associated with money transfer. Finally, agents with $\alpha = 0$ obtain increased gains from trade primarily through higher money transfer from 0.575 ticks in consolidated markets to 0.738 in fragmented markets.

As discussed earlier, traders with $\alpha = 0$ are more likely to be liquidity suppliers. Therefore, in a consolidated market they have to place more aggressive orders to jump ahead of existing limit order queues to increase execution probability. This increases price competition due to the consolidation of all order flow in a single trading venue. On the contrary, in fragmented markets, traders do not necessarily have to submit an aggressive order for raising their probability of execution. Instead they can choose the second market to jump ahead of the outstanding limit orders in the first market. This reduces price competition in each market and consequently agents with $\alpha = 0$ obtain better terms of trade.[14] On the other hand, agents with higher absolute valuations for the asset act as liquidity consumers and bear the increased cost associated with reduced price competition in fragmented markets. This is because liquidity providers are willing to submit more limit orders due to the possibility of capturing higher spreads. This leads to an increase in consolidated depth across the two limit order books inducing intrinsically motivated traders to submit market orders and reduce waiting costs. The net outcome is a slight reduction in overall welfare.

3.3 Liquidity and Price Efficiency

In this section we compare different market quality measures in consolidated versus fragmented markets. Table 3 reports the following liquidity measures: quoted bid-ask spread, effective spreads, average number of limit orders at the ask quote (total and effective traded), average number of limit orders on the sell side of the book (total and effective traded) and microstructure noise. The quoted spread is calculated observing the market every 10 units of time whenever bid and ask exist. The effective spread is calculated with all transactions as mean of $|p - m|$,

[14] The absence of time priority across markets is a pre-requisite for this result.

where p is the transaction price, and m is the midpoint between the bid and ask quotes. Since the model is symmetric we show liquidity measures for one of two markets, as both markets exhibit the same features.

Observation 3. *Fragmented markets impair liquidity in each local market. This is reflected in wider spreads and lower market depth. Despite liquidity reduction in local markets, consolidated depth is also higher in fragmented markets which represent a positive effect of market fragmentation.*

As Table 3 reports, in a consolidated market, the quoted spread is 1.62 ticks and the effective spread is 0.715 ticks, while in fragmented markets these measures increase to 3.121 and 1.204 ticks, respectively. These two results suggest that liquidity in local markets is lower when another venue competes for order flow. This can be explained by shifts in orders' aggressiveness; when a consolidated market absorbs all the order flow, liquidity suppliers have to place aggressive orders to improve their probabilities of execution or wait in the queue respecting price and time priority, which entails narrower spreads. This effect is consistent with Gajewski and Gresse (2007), Bennett and Wei (2006), and Amihud et al. (2003). In contrast, orders can execute at worse prices in fragmented markets because agents can circumvent time priority in one market by placing an order in the second one without having to become more aggressive. This is reflected in wider spreads.

Table 3 also reports a higher number of limit orders (or depth) at the best ask and a higher number of limit orders overall on the sell side of the book. Depth at the best ask is reduced from 1.487 in consolidated markets to 0.942 in fragmented markets. This is because part of the order flow executes against limit orders posted in one market and the rest in the other market. Therefore, other thing being equal, execution probabilities are smaller for a single market and, as a consequence, each market attracts less limits orders in comparison when it operates alone. van Kervel (2015) show that liquidity providers duplicate their limit orders across multiple markets. However, traders in our model cannot adopt this strategy because they have a single share to trade, thus liquidity provision fragments between two markets. These results are suggestive of worse liquidity in each limit order book when markets are fragmented.

However, we also observe an improvement in aggregated consolidated depth. For example, consolidated depth at the best ask, which is simple twice the depth in a single market, is $2 \cdot 0.942 = 1.884 > 1.487$. This result is consistent with Foucault and Menkveld (2008) who study competition between two pure limit order markets and observe a similar effect on consolidated depth. Besides, Degryse et al. (2015) find consistent empirical evidence for Dutch Stocks and concludes that more lit fragmentation entails higher consolidated depth but lower local depth. These results suggest that traders who do not or cannot access all trading venues might be adversely affected, because they face a less liquid market.

Observation 4. *Microstructure noise is higher in fragmented markets.*

Microstructure noise is generally related to frictions in financial markets that induce temporary deviations between the transaction price p_t and the fundamental value v_t. If we describe the transaction price as the fundamental value v_t plus microstructure noise ξ_t, i.e., $p_t = v_t + \xi_t$. In Table 3, we report the absolute mean and standard deviation of microstructure noise. We observe that both, absolute mean and standard deviation of microstructure noise increase in fragmented markets. For example, the standard deviation increases from 0.866 ticks in a consolidated market to 1.061 ticks in fragmented markets. This suggests that price volatility unrelated to fundamentals is positively associated with market fragmentation and less information is impounded in prices as in Madhavan (1995).

4. Empirical Predictions

In this section, we generate the empirical predictions that support the theoretical results obtained in the previous section.

Empirical Prediction 1. *After consolidation trading volume does not decrease.*

This prediction supports Observation 2(i). In fragmented markets there exist agents engaging in arbitraging mispricings between the different venues. These agents have no intrinsic motives to trade. Thus their trading activity is eliminated when order books are consolidated. If trading volume does not decrease after consolidation this suggests that other traders (with intrinsic reasons to trade) enter the market or are now more active. New traders with intrinsic reasons can participate gainfully due to improved liquidity. These traders earn their private values thus leading to an overall increase in welfare. This relationship between gains from trade and trading volume is subject to the caveat that waiting costs do not increase disproportionately relative to the additional gross trading gains. In Table 2, we observe that, while waiting costs are higher in consolidated markets, the difference is small relative to the private values.

Empirical Prediction 2. *(i) Quoted bid-ask spreads after consolidation are lower than quoted local spreads in the pre-event period. (ii) Quoted depth is higher than the fragmented quoted local depth after consolidation.*

This prediction corresponds directly to Observation 3.

Empirical Prediction 3. *Realized spreads decrease after consolidation.*

This prediction supports Observation 2(ii). A shift in welfare gains from agents with no intrinsic value to those trading for intrinsic reasons should lead to a decrease in realized spreads.

Empirical Prediction 4. *Price efficiency, measured by absolute values of autocorrelation and variance ratios, is improved after consolidation*

This prediction corresponds to Observation 4.

5. Sample, Data, and Summary Statistics

In this section we describe NYSE Euronext's institutional arrangements leading up to the introduction of the Single Order Book. We also describe the construction of the stock sample as well as the data used in our analysis. We end this section by summarizing the market activity for the stocks.

5.1 Introduction of Euronext Single Order Book

Stock listings on NYSE Euronext pertain to a listing on any of the individual markets.[15] Until 13 January 2009, each listing corresponded to the operation of one order book. For example, a stock listed on Euronext Paris would be traded on the limit order book of Euronext Paris. However, some companies had multiple listings in different Euronext markets and, hence, trading in these stocks would take place in multiple parallel Euronext order books, besides other competing markets. As the rules and trading protocols governing individual Euronext order books were identical, trading in multiple order books led to increased fragmentation without enhancing overall competition between different market operators. On 16 August 2007, the exchange announced its intention to eliminate this arrangement for their Paris, Amsterdam and Brussels markets by unifying all trading in these markets onto a single order book, the so-called "Market of Reference" (MoR). This new arrangement was implemented on 14 January 2009, reducing the level of fragmentation for a select group of stocks. The competitive position between the different market operators remained unchanged.[16]

[15] Euronext was formed in 2000 as the result of a merger of Paris Bourse, Amsterdam Stock Exchange and Brussels Stock Exchange. Later, in 2002, the Lisbon Stock Exchange became the fourth exchange to merge with Euronext. Since then the exchange operates as a single, pan-European platform for trading securities listed on all the four exchanges. In 2007, Euronext merged with the NYSE to form NYSE Euronext, which was taken over by Intercontinental Exchange in 2012. In 2014, Euronext was spun off through an IPO.

[16] Euronext, in its press release announcing the event, made clear that the trading environment was unchanged: "The Single Order Book will have no impact on the NSC system as the market rules and order book management

5.2 Sample Selection and Estimation Approaches

A total of 45 instruments, cross-listed on at least two of the three Euronext markets, are affected by the event. However, we reduce the sample of treated stocks that we use in our study for several reasons. First, we remove stocks whose primary listing is not on Euronext. These include stocks whose main trading activity takes place in other European markets or in the United States. Second, we eliminate exchange-traded mutual funds since we do not expect their trading activity to be comparable to that of listed companies. Finally, we require that there not only exist multiple Euronext order books before the event, but also that the total share of trading activity on the less active order books is at least equal to 1% of total Euronext trading volume. This reduces the list of instruments to ten. We further exclude one additional stock due to data errors reducing our final sample to nine stocks.[17] The number of stocks is small due to the unique nature of the event we study. Nonetheless, our sample consists of the whole population of stocks affected by the event, except a subset of stocks which are excluded through an objective criterion.

We construct a matched control group of stocks based on stock price and market capitalization obtained from Compustat Global. Specifically, for each stock in our treatment group, we identify the stock which is closest to it in terms of these two criteria as on the last trading day of 2008 (30 December 2008). The population of stocks from which the control group is constructed comprises all stocks which have a primary listing on NYSE Euronext, and which are not affected by the event.

Davies and Kim (2009) simulate the matching performance of a control group constructed using multiple criteria such as price volatility, trading volume and industry classification, in tests of differences for variables typically used in the microstructure literature, and conclude that one-to-one sampling without replacement based on stock price and market capitalization provides the best results. They also show that results obtained by matching based on the distance metric employed by Huang and Stoll (1996), and subsequently in many other market microstructure studies, are similar to those obtained when using the Mahalanobis distance measure. For robustness purposes we use both distance measures. We use the natural logarithms of the price and market capitalization when measuring the Mahalanobis distance because close

will remain unchanged [...] In practice, from a trading perspective, Single Order Book implementation simply means the end of order book trading on marketplaces other than the market of reference."

[17]One stock in our sample was listed in all three Euronext markets. However, we exclude the least active limit order book as it had a market share of 0.3%.

matches should be characterized by small distances in relative rather than absolute terms.

Using a control sample allows us to identify the effects of reduced fragmentation, implicitly controlling for market-wide changes in variables such as liquidity and volatility. It also allows us to control for two additional market-wide changes that were implemented by Euronext around the same time as the introduction of a single order book. First, a harmonized settlement platform for all French, Dutch and Belgian stocks was implemented on 19 January 2009.[18] Second, an improved trading platform, the Universal Trading Platform, was introduced on 16 February 2009.[19] Both of these changes were market-wide events affecting both the control and treatment stocks. Consequently, we can attribute any difference in trading activity and market quality between the two groups to the market consolidation event.

For the purpose of our analysis, we define all days from the beginning of December 2008 to 13 January 2009 as the pre-event period and all days from 26 January 2009 to the end of February 2009 as the post-event period. We exclude eight trading days from the event date until the end of the subsequent calendar week in order to eliminate any effect associated with the transition.

5.3 Data and Variables

We use high-frequency data from Thomson Reuters Tick History (TRTH) between December 2008 and February 2009 for the purpose of our analysis. This data contains trades and order book updates time-stamped with a millisecond resolution. We apply several filters to the data. First we eliminate all order book updates where the best bid or ask prices are zero, or the bid-ask spread is negative. We also exclude trades which are not executed during the continuous trading sessions and which have a time-stamp before 09:01 or after 17:29.

We compute several variables capturing the trading activity, liquidity and price efficiency of the stocks. First, we compute daily trading volume and number of trades. We capture daily quoted liquidity by time-weighted quoted spreads and time-weighted top-of-book depth. These measures are computed for the more active market i.e., the one having higher trading volume during the pre-event period, and based on the inside quotes i.e., the highest bid and the lowest ask across the two limit order books. Next, we compute trade-weighted effective spreads as a measure of realized liquidity. These capture the actual transaction costs incurred by market orders taking into account the presence of hidden liquidity and limited top-of-book depth. We

[18] This was known as the Euroclear Settlement for Euronext-zone Securities (ESES)

[19] The new trading system had "superior functionality, faster speed and much greater capacity."

further decompose effective spreads into price impact and realized spreads using mid-quotes 10, 30 and 60 seconds in the future. The former capture the level of information in a trade and the latter capture liquidity providers' compensation after accounting for adverse selection losses associated with informed orders. We use future inside mid-quotes as well as local mid-quotes while computing realized spreads and price impacts. Measuring effective spreads and their decomposition for the local market as well as across the two limit order books in the pre-event period allows us to examine the impact of market consolidation perceived by investors with and without access to one or both markets respectively. This is critical as theory sometimes predicts opposite impact of fragmentation on local liquidity and global liquidity. Needless to say, the two variants of liquidity are exactly identical for the control stocks as well as for the treatment stocks after the introduction of the single order book.

In addition to the liquidity measures, we construct two price efficiency measures: return autocorrelations and variance ratios. Return autocorrelations are measured at 10 second, 30 second, 1 minute and 5 minute intervals. Variance ratios capture the deviation between long-term and short-term return variance and are calculated as one minus the ratio of long-term and short-term return variance, each scaled by the respective time periods. As in Boehmer and Kelley (2009), we report absolute values of both measures as we are interested in departures from a random walk in either direction. Similar to the liquidity measures, we calculate price efficiency for the most active market and the inside quotes. We calculate variance ratios between 10 and 30 seconds, 10 seconds and 1 minute, 10 seconds and 5 minutes, 30 seconds and 1 minute, 30 seconds and 5 minutes, and 1 and 5 minutes returns variances. If prices follow a random walk, both measures should have a value of zero. We report the results return autocorrelations at 30 second and 5 minute intervals, and for variance ratios based on 30 second and 5 minute return variances. The results for other frequencies are consistent with those reported in the paper.[20]

5.4 Summary Statistics

Table 4 describes the characteristics of stocks in the treatment and Huang and Stoll (1996) control group. The average market capitalization across stocks in the treatment (control) group was €4.4 (€4.8) billion and the average stock price was Euro 18.4 (18.2), suggesting that the quality of the match based on these two variables is very high. Differences between treatment

[20] These results are available upon request.

and control group are naturally larger for other variables than those that were used for the matching.[21] For example, the price volatility (not reported) of an average stock in the treatment group is higher than that of an average stock in the control group. Quoted spreads for stocks in the control group are high while top-of-book depth is low. This is one reason why we run all our analyses both for levels as well as logarithms of our variables of interest.

The share of the more active venue as a percentage of total Euronext volume across all the days before the event ranges from 54% to 98% across the nine stocks in the treatment group. The simple (volume-weighted) average across all stocks is 78% (62%). This suggests that almost 40% of the total Euronext volume was executed on the less active market. The market share of the sole listing Euronext venue for the stocks in the control group is, by construction, 100%. Trading activity when measured in terms of number of trades also provides a similar picture.

Table 5 provides the pre-event and post-event mean values of the different variables for the stocks in the treatment and control groups. We separately report these statistics for the full sample of nine stocks and for the four largest stocks. A stock in the treatment group generates an average daily trading volume of €28 million as compared to €12 million for the control group. Quoted liquidity is more similar between stocks in the treatment and control groups. The more active order book has a bid-ask spread of 1.6% and depth of €13,258. The corresponding measures for the control stocks are 1.3% and €15,880 respectively. Quoted liquidity measured at the inside quotes though higher than the treatment stocks. Effective spreads again are higher for the treatment stocks with a mean of 1.9% as compared to 1.2% for the control group. This increase seems to be entirely attributable to higher realized spreads for the treatment stocks; realized spreads measured at 60 seconds are 1.5% and 0.8% for the treatment and control stocks respectively. Price efficiency seems to be similar for both groups with absolute autocorrelations between 0.06 and 0.10, and variance ratios around 0.2. Based on the summary statistics it seems as though trading activity and liquidity increased after the introduction of single order book.

6. Results

The summary statistics discussed in the previous section point to significant improvements in market quality. However, it is not clear whether these improvements are statistically significant

[21] This is not problematic according to Davies and Kim (2009) and necessary so as to limit the dimensions used to define matches.

and whether their magnitudes remain economically significant after controlling for market-wide shifts in trading activity, liquidity and price efficiency. We use two alternative estimation approaches to answer these questions. First, we estimate a standard panel difference-in-difference regression with stock and day fixed effects and standard errors double clustered by stock and day. In the second estimation approach, we compute mean levels of the measures of interest for each stock in the pre- and post-event periods under consideration. We pair the changes from pre- to post-event period for treatment stocks and their controls and use the matched samples to conduct pairwise Student t-tests and Wilcoxon signed rank tests. We apply both estimation approaches for levels and logs of the variables of interest. Davies and Kim (2009) suggest the use of the Wilcoxon signed-rank test for reasons of statistical power. On the other hand, we expect that the small number of observations in our sample will result in rather limited statistical power not just for the non-parametric tests but also for the parametric Student t-tests. We report the results from the panel difference-in-difference estimation.[22] Subsection 6.1 examines the reduction in arbitrage activity after the introduction of a single order book. This is followed by Subsection 6.2 considers the effect on trading volume and liquidity. Finally, we conclude this section by analyzing the price efficiency results in 6.3.

6.1 Arbitrage

Consolidation of trading into a single order book should lead to reduction in arbitrage activity between the two order books operated by Euronext. In order to identify the magnitude of such arbitrage trades we start by first identifying instances of a crossed order book. This happens when the best bid on one order book is higher than the best ask on the second order book. Such a crossed market can arise as a result of new order(s) submitted to either or both order book(s). Next, we identify whether these instances are resolved through a trade, quote-update, or both. This approach is similar to Foucault et al. (2014) who define resolution through trades as "toxic arbitrage" if the following two conditions are fulfilled: (1) prices offered in different markets allow aggressive traders to earn a profit by trading against the bid on one market and ask on the other; (2) they are able to do so because of a liquidity providers' slow reaction to new information, rather than because of liquidity providers offering attractive prices to manage their inventories. Fragmentation is an obvious precondition for arbitrage trades to occur. As such, if consolidation reduces toxic arbitrage trades, market-makers can offer less expensive

[22] The results from the Student t-tests and Wilcoxon signed ranked tests are available upon request.

liquidity to other market participants reflecting in reduced realized spreads. For each stock-day, we calculate the number of unique crossed instances, percent of day when inside spreads are negative, the number of instances when inside spread is one, two, three, four, and five and above ticks, and total trading volume contributing to the resolution of crossed market. Table 6 reports the average values for each stock across all days in the pre-event period.

The frequency of unique instances of a crossed market for an average day range from 0.3 to 622 across all stocks, with an average value of 124 which represents one instance every 4 minutes. However, the four liquid stocks have a much higher number of instances probably as these stocks have relatively smaller spreads to begin with. ISPA, the largest stock in the sample, experiences a crossed market once every 49 seconds. An average stock has a negative inside spread for 6.4% of the continuous trading session, with larger stocks again reporting a much higher percentage. The magnitude of negative spreads is less than three ticks in 68% of the instances and is five or more ticks in 21% of the instances. However, a negative inside spread of one tick can be substantial for a low priced stock. For example, FOR which is priced at €0.9 has a minimum tick size of €0.001 or 11bps. Finally, 7.8% of the total trading volume on Euronext for an average stock can be attributed to the resolution of instances where the two markets are crossed. However, this ranges from 4.1% for GLPG to 14.1% for THEB.

6.2 Trading Volume and Liquidity

Table 7 provides the results of the difference-in-difference regression for trading volume, quoted spread, and quoted top-of-book depth. Total volume is likely affected by the market consolidation event through several channels. First and foremost, the presence of multiple markets requires some agents who make sure that prices in these markets are consistent with each other by arbitraging inefficiencies. In today's markets, this task is generally performed by high-frequency traders who use high speed access to multiple markets for this purpose (see Budish et al. (2015) for example). With consolidation such arbitrage opportunities within Euronext will be eliminated resulting in a drop in trading volume.[23] We examined this issue in the previous section. On the other hand, liquidity in the consolidated market is also likely to be positively associated with trading volume. First, low trading costs allow more market participants to enter the market and gainfully execute their strategies as in Pagano (1989) and Chowdhry and Nanda (1991). Second, to the extent alternative trading venues are substitutes, improvements

[23]However, overall arbitrage opportunities will not be completely eliminated due to the presence of competitors such as BATS, Chi-X, and Turquoise that trade some stocks in our sample.

in liquidity on one trading venue will lead to migration of volume away from other trading venues. The coefficient for level trading volume is €3.6 million and significant at 10%. The coefficient for log trading volume as well as for level and log number of trades is also positive but insignificant. This confirms the Empirical Prediction 1.

Next, we observe that the introduction of a single order book led to a statistically and economically significant reduction in quoted bid-ask spreads. Specifically, the more active of the two Euronext markets experienced a reduction in spreads of 81bps or approximately 30% relative to the control group. This change is also statistically significant. The improvements seem to be more prominent for the less liquid firms considering the coefficients for the four largest stocks are insignificant. Interestingly, changes in inside spreads appear to be insignificant for the full sample but positive and significant for the large firms. This is likely because the introduction of a single order book eliminates instances of crossed inside quotes leading to a mechanical increase in inside spreads, especially for the more liquid and low priced stocks. This confirms Empirical Prediction 2(i). For large stocks, the results are in the opposite direction, though small and insignificant.

When the same intermediaries are providing liquidity across multiple markets, it may be that they duplicate their liquidity on display, accepting the risk of facing trades on both order books. Sophisticated liquidity providers such as high-frequency traders are known to quote in multiple markets but immediately cancel their limit orders on one market after their order on the other is hit (See van Kervel (2015) for example). In this case while aggregate liquidity across the two markets may be higher, it can remain inaccessible to market participants who do not have a speed advantage. On the other hand, if liquidity providers consider the risk of being hit on both order books simultaneously to be significant, they may decide to offer a smaller amount of shares on each market than they would in a single market. In this case, their depth on the discontinued order book will migrate to the surviving order book, leading to an increase in local depth while keeping aggregate depth unchanged. We find that top-of-book depth for the more active market and at the inside quotes increases between €4,500 and €5,000 after the introduction of a single order book although this increase is not significant. However, for the four largest stocks these measures increase by approximately €12,000 or almost 50%. This suggests that the results for the largest stocks are inconsistent with van Kervel (2015). One alternative explanation is that as best quotes in fragmented are less likely to be exactly equal, the depths simply do not add up when measured relative to the inside quotes. This confirms

Empirical Prediction 2(ii) for the largest stocks. For the full sample, although the results are insignificant, the signs of the coefficients are in the right direction.

Theory suggests that consolidation of trading leads to improvements in liquidity due to positive network externalities. These externalities manifest themselves through several channels. First, due to higher concentration of natural liquidity, market-makers' inventory holding costs decrease. Second, their monitoring costs also decrease and along with it the risk of stale quotes being picked off decrease. Third, increase in trading volume allows liquidity providers to allocate fixed costs over a larger number of trades.[24] These cost benefits should translate into lower realized spreads. On the other hand, market consolidation will affect price impact ambiguously. On the one hand, improved price efficiency and reduction in arbitrage activity should eliminate less informed trades leading to an increase in per-trade price impact. On the other hand, lower transaction costs and higher price efficiency will likely attract more investors to the market. This clientele effect can increase or decrease price impact depending on whether new investors are more or less informed relative to the existing investor base. Table 8 provides the results for effective spread and its decomposition into realized spread and price impact. Effective inside (local) spreads decreased by 93bps (139bps) or 14.5% (37.5%) after the introduction of a single order book. The changes in level spreads are significant at 10% and those in log local spreads at 5%. Consistent with the network effects described above, realized inside spreads at 10, 30, and 60 seconds decreased by 37.6%, 54.7%, and 61.5% respectively and are also statistically significant. The corresponding decrease in realized local spreads is larger in magnitude and again statistically significant. Level price impacts, although also decreasing, were always lower in magnitude than realized spreads and only significant for the local markets. This suggests that, for the whole sample, reduced market-makers' costs after the introduction of a single order book were largely passed on to liquidity consumers due to improved competition between limit order traders. For the four largest stocks, we observe a 27.1%, 24.2%, and 21.5% increase in price impact measured at 10, 30, and 60 seconds which leads to a weakly significant increase in effective spreads. Realized spreads remain unchanged possibly due to already high levels competition between limit orders even in fragmented markets. Overall, these results point to improvements in liquidity due to market consolidation, especially for small stocks. This confirms Empirical Prediction 3.

[24] The model does not incorporate the impact of fixed costs.

6.3 Price Efficiency

Information efficiency may also be affected by the fragmentation of markets. If an asset is traded on multiple markets, small price dislocations can occur, making prices on each book less efficient than they would be if all demand and supply were to meet on a single order book. Table 9 presents the results. We find that the variance ratio becomes closer to one after the implementation of the single order book, however the change is statistically insignificant. The results for autocorrelations also point to improved price efficiency although only 5-minute autocorrelation is significant at 10%. The results for the four largest stocks while in the same direction are always insignificant. This points to improvements in price efficiency only being restricted to the smaller stocks. This weakly confirms Empirical Prediction 4.

7. Conclusion

We examine the impact of market fragmentation on various aspects of market quality. Our theoretical model and the unique characteristics of the event we exploit in the empirical part of the paper allow us to separately evaluate the impact of a reduction in market fragmentation while keeping the level of competition in the market unchanged. Existing empirical research on this question necessarily investigates the joint impact of changes in fragmentation and competition as the two generally go hand-in-hand. However, the recent wave of mergers between different exchange operators prompted largely as a result of competition from different kinds of lit and dark markets has highlighted the need to separately examine the effect of these two forces. When exchanges merge, competition between individual exchanges belonging to the same exchange operator is likely to be minimal, if not non-existent. This leads to a reduction in the level of competition while overall order flow fragmentation remains unchanged. Our results are relevant in light of this issue.

We find that liquidity and short-term price efficiency generally improve as a result of the consolidation of order flow. These results are qualitatively similar for data generated from the numerical simulation of optimizing market participants, and for the empirical observed data surrounding the consolidation of Euronext's limit order books. The results suggest that positive externalities associated with consolidating order flow in a single location (or fewer locations) still exist and are substantial. This is true even in modern electronic limit order markets where the activities of high-frequency traders serve to integrate fragmented order books. For an investor without the technological ability to choose to trade on the most advantageous venue,[25] a consolidation in order flow, at least between non-competing markets, will likely lead to improvements through reduced transaction costs.

Our results also have important policy implications. Regulators could significantly improve the allocative efficiency of markets by preventing individual market operators from artificially keeping a high level of order flow fragmentation without providing meaningful benefits to investors through market differentiation. Market operators, in turn, could also provide their customers substantial savings in transaction costs by consolidating the order flow from similarly-structured multiple order books.

[25] As a result of latency, no investor truly possesses this ability, even though we do not make this assumption in our theory.

Appendix A. Solving for Equilibrium

A.1 Numerical Parameterization

For our numeric simulations, we set parameters based on the existing literature. In single market scenarios, we follow parameters proposed by Goettler et al. (2009), while in multi markets, we adjust some parameters to make both scenarios properly comparable. The multi market scenario involves a higher levels of complexity in comparison with the single market. First, the state space is massively increased if we consider two markets instead of one. Second, we need to track the evolution of two limit order books in continuous time, which represents an additional difficulty for that scenario.

Single Market Parameterization: We use the following parameters to generate the single market equilibrium.

- Agents can only trade in the book $L_{1,t}$.
- We normalize the rate of the Poisson process for new trader arrivals, λ, to one. This means that a unit of time in our simulations represent the average time between new trader arrivals. On average, a trader reenters the market after four units of time, thus the rate of the Poisson process for reentries is, $\lambda_r = \frac{1}{4}$. This reentry rate is assumed constant across all traders.
- One tick in the simulation corresponds to one-eighth of a euro.
- We assume that the distribution of private value F_α is discrete with support $\{-8, -4, 0, 4, 8\}$ and cumulative distribution $\{0.15, 0.35, 0.65, 0.85, 1.0\}$. This distribution is based on Hollifield et al. (2006) who estimate the distributions of private values for three stocks on the Vancouver exchange.
- Innovations in the fundamental value of the asset, v, occur according to a Poisson process at rate $\lambda_v = \frac{1}{8}$, which means that the fundamental value jumps, on average every eight units of model time. Whenever v changes, it increases or decreases in a fixed amount $f = 1$, each with the same probability.
- We assume $\rho = 0.05$. We used lower values and found results qualitatively similar.
- When a trader submits a limit order at time t, the price is restricted in the range $[v_t - k, v_t + k] \cap \mathcal{P}$, where $\mathcal{P}$ is the set of feasible prices. This assumption is made for computational

tractability, since we need a finite set of prices. The amount k is chosen large enough that it does not affect the equilibrium. For the simulations we choose $k = 6$.[26]

- The market is transparent: each trader can observe quotes, the depth at the quotes, aggregate depth on both sides of the market, and the last transaction, which includes the price, direction and book for that order.

- We set the cancellation fee to zero. This allow us to explore the trading process with less frictions.

Multi Markets Parameterization: We use the following parameters to generate the multi market equilibrium.

- We allow agents to trade in the books $L_{1,t}$ and $L_{2,t}$

- We set the arrival rate $\lambda = 2$ to inject liquidity with the same intensity to each venue in comparison with single market.

- All market parameters such as tick size, cancellation fee, and transparency are set identically in the two markets allowing a comparison of the results to a single market scenario.

A.2 Details of the Numerical Algorithm

We make further assumptions to allow our simulations to be computationally tractable. First, the fundamental value evolves over time, thus the set of feasible prices is large (although finite as our simulations are finite). However, agents only care about the difference between the transaction price and the fundamental value v_t, so we normalize prices (both current and historical) relatively to v_t, significantly reducing the state space in our numerical simulations. Ideally, we would like to condition agents' strategies on all information available in the order books, but this is computationally intractable. Instead, we restrict the state space for a trader as follows. Let s_t denote the state observed by an agent at the time t, then:

$$s_t = \{L_{1,t}, L_{2,t}, \delta_t, v_t, \alpha, \rho, \tilde{a}_t\} \tag{A.1}$$

where $L_{m,t}, m \in \{1, 2\}$ denotes the state of order book m at time t including bid and ask prices $(B_{m,t}, A_{m,t})$, depths at these prices $(l^B_{m,t}, l^A_{m,t})$, and aggregate depth on buy and sell side of the

[26] We experimented with higher values of k and noticed that traders rarely send orders so far from the fundamental value.

market $\sum\{l^i_{m,t} > 0\}$ and $\sum\{l^i_{m,t} < 0\}$. δ_t indicates the most recent transaction including the book where it was executed, price and direction, v_t is the fundamental value, α and ρ are agents' private value and delay rate respectively, and $\tilde{a} = (\tilde{b}, \tilde{p}, \tilde{x}, \tilde{q})$ is the state of his previous order.

We consider three exogenous events:

- Changes in the fundamental value: Every time the fundamental value v changes, it increases or decreases with by f with probability $\frac{1}{2}$, leading to a shift in the position of all standing limit orders relative to the fundamental value.
- New trader arrivals: Every time a new trader arrives to the market, he observes the current state s and takes the optimal action a^*. If a^* is a market order, he executes and leaves the market forever. If he submits any other order we draw a random time for his re-entry.
- Re-entry of 'old' traders who have not yet executed: Every time a trader returns to the market, he observes the current state s and chooses an optimal action. If the order he selects is a market order, he executes and leaves the market forever. If he submits any other order we draw a new return time for his re-entry.

A.3 Learning Process

At time t, each action a in the state s represents the agent's current belief with an associated expected payoff $U_t(a|s)$. Consequently, the agent selects the action a^* in the state s at time t if and only if $a^* \in \underset{a \in \Gamma(s)}{\operatorname{argmax}}\, U_t(a|s)$. In this case, the optimal value of state s from the Bellman equation for the traders' dynamic maximization problem is given by $V(s) = U_t(a^*|s)$. For each pair (a, s), there is an initial belief $U_0(a|s)$. Note that as any $U_0(\cdot)$ can lead to an equilibrium, we choose it choose a value enabling us to quickly reach the equilibrium, similar to Goettler et al. (2009). These initial beliefs are updated as the algorithm progresses. Suppose the optimal action a^* is not a market order, i.e., it is either a limit order or no order, and further suppose that at some future time t' the trader re-enters and observes a state s' with an optimal value $V(s')$. Then, the belief $U_t(a^*|s)$ is updated as:

$$U_{t'}(a^*|s) = \frac{n}{n+1} U_t(a^*|s) + \frac{1}{n+1} e^{-\rho(t'-t)} V(s') \tag{A.2}$$

where $n = n(a^*, s)$ is a positive integer that is incremented by one each time action a^* is chosen in state s. Similar to Goettler et al. (2009), we periodically reset n to n_0 during the simulation to obtain quicker convergence.

Similarly, suppose an agent submits a limit sell (denoted by a^*) when he faces the state s at time t, and this order executes against a market order submitted by another trader at time t'. In this case we update the belief $U_t(a^*|s)$ with the realized payoff associated with the transaction as:

$$U_{t'}(a^*|s) = \frac{n}{n+1}U_t(a^*|s) + \frac{1}{n+1}e^{-\rho(t'-t)}x(\alpha + v_{t'} - p) \tag{A.3}$$

In the simulations, if all agents take the optimal action given their current beliefs, there is a possibility that the algorithm reaches a sub-optimal equilibrium, maybe because traders have not learned payoffs of other actions. To ensure that beliefs are updated for all actions in every state, with a small probability ϵ a trader trembles over all suboptimal limits orders with same probability, which allows the trader to visit states that he never would have been able to visit without trembles.

A.4 Convergence Criteria

The simulation has three main phases:

(i) In the first phase, we update the beliefs of agents for sufficient time to ensure equilibrium is reached (two billion of new arrivals for single market scenarios and five billion for multi market).

(ii) After the first phase, we verify that the algorithm has converged properly.

(iii) The algorithm reaches the equilibrium if the convergence criterion is satisfied. We then fix the belief of agents (i.e., there is not more learning process), disallow trembles (i.e., $\epsilon = 0$) and simulate the model for another 300 million of new arrivals to obtain our results in equilibrium.

In (ii), we use a convergence criterion similar to Goettler et al. (2009). We take a snapshot of beliefs at the end of (i) and simulate several millions of additional new arrivals. We then take another snapshot of beliefs and, intuitively, if both snapshots are sufficiently similar, the algorithm has converged.

Formally, we evaluate the weighted relative difference between the beliefs at the end of (i), $U_{t_1}(\cdot)$ and the belief at the end of the current 600 million new trader arrivals, $U_{t_2}(\cdot)$. We require the weighted relative difference not to exceed 1%, which can be expressed as:

$$\sum_{(a,s)\in\mathcal{X}} \frac{U_{t_1}^{k_1}(a|s) - U_{t_2}^{k_2}(a|s)}{(k_2 - k_1)U_{t_1}^{k_1}(a|s)} < 0.01 \tag{A.4}$$

where $\mathcal{X}$ is the set of all actions a selected in the state s during the first phase, k_1 is the number of times the action a has been taken in state s at the end of (i) and $k_2 \geq k_1$ is the number of times it has been chosen at the end of current 600 millions of new traders arrivals.

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Table 1. Strategies by Trader Type

This table reports the proportion of market orders submitted by each trader type as percentage of all market orders submitted, the proportion of limit orders submitted by each trader type as percentage of all limit orders submitted, and proportion of non order by each trader type as percentage of all traders that choose not an order in equilibrium. All percentages are determined as mean of 20 million market new entries in equilibrium. Standard errors for all trader strategies are small enough since we use a large number of simulated events. The Markov equilibrium is obtained independently for each scenario.

Scenario	Order Type	Private Value —α—		
		0	4	8
Single Market	Limit Orders	67.9%	26.1%	6.1%
	Market Orders	13.4%	38.8%	47.7%
	Non Orders	80.3%	15.5%	4.2%
Multi Markets	Limit Orders	68.0%	26.8%	5.2%
	Market Orders	5.8%	43.4%	50.9%
	Non Orders	74.8%	21.7%	3.5%

Table 2. Decomposition of Payoffs by Trader Type

Average payoff, waiting cost and money transfer per trader differentiated by private value (all three measures in ticks). Payoffs determined are as in Table 1, while waiting costs and money transfers are described equation (4). The first row reports single market scenario. In the second we report multi markets scenario. The average payoffs, waiting cost and money transfer per trader are determined as mean of payoffs over 20 million market new entries in equilibrium. Standard errors for measures are small enough since we use a large number of simulated events. The Markov equilibrium is obtained independently for each scenario.

Scenario	Average payoff per trader				Waiting cost per trader				Money transfer per trader			
	Private Value —α— 0	4	8	Total	Private Value —α— 0	4	8	Total	Private Value —α— 0	4	8	Total
Single Market	0.575	3.497	7.236	3.742	0.000	-0.359	-0.173	-0.195	0.575	-0.144	-0.591	-0.062
Multi Market	0.738	3.511	6.984	3.721	0.000	-0.338	-0.127	-0.173	0.738	-0.151	-0.889	-0.106

Table 3. Market Quality Measures: Consolidated Versus Fragmented Markets

This table shows different market quality measures for one of the two markets such as Bid-Ask Spread, Effective Spread, Number of limit orders at the ask (total and effectively traded), Number of limit orders on the sell side of the book (total and effectively traded) and absolute mean and standard deviation of microstructure noise. Each column is a different scenario, the first represent single market and the second, multi markets. The mean and standard deviation of microstructure noise are calculated with transactions as transaction price (p_t) minus the fundamental value (v_t) in ticks. All market quality measures are determined as mean of 20 million market new entries in equilibrium for one of the two market. Since both markets have identical characteristic, we do not need to report all measures for the second market. Standard errors for all Market quality measures are small enough since we use a large number of simulated events. The Markov equilibrium is obtained independently for each scenario.

	Single Market	Fragmented Markets
Bid-ask spread	1.620	3.121
Effective spread	0.715	1.204
No. of limit orders at the best ask	1.487	0.942
No. of limit orders at the ask (effectively traded)	0.586	0.525
No. of limit orders on the sell side of the book	2.595	1.392
N. of limit orders on the sell side of the book (effectively traded)	0.823	0.598
Microstructure noise: Mean $\|v_t - p_t\|$	0.662	0.848
Microstructure noise: Std. Dev. $(v_t - p_t)$	0.866	1.061

Table 4. Stock Characteristics

This table reports the characteristics of the treatment stocks and the corresponding control stocks generated based on Huang and Stoll (1996). Market Capitalization is the product of shares outstanding and Stock Price as on 31 December 2008, Trading Volume and Number of Trades is the average daily trading volume and number of trades for each stock between 1 December 2008 and 13 January 2009. We also report the market share of the two Euronext order books. Large (Small) order book is the order book with higher (lower) trading volume.

Panel A: Treatment Stocks

	Market Cap	Stock	Trading Volume			Number of Trades		
	€ million	Price (€)	€ '000	% Large	% Small	Count	% Large	% Small
DEXI	3,355	2.9	8,514	54.3%	45.7%	2,843.1	52.6%	47.4%
FOR	2,187	0.9	25,239	71.5%	28.5%	6,613.4	71.7%	28.3%
ISPA	24,985	17.2	180,346	55.6%	44.4%	18,231.8	52.7%	47.3%
UNBP	8,598	104.9	37,687	87.7%	12.3%	4,268.1	86.4%	13.6%
GLPG	80	3.8	183	77.1%	22.9%	67.9	27.9%	72.1%
ONCOB	87	6.6	23	90.7%	9.3%	3.5	73.5%	26.5%
RCUS	193	6.2	119	98.0%	2.0%	68.3	98.5%	1.5%
VRKP	105	20	43	94.4%	5.6%	19.5	94.3%	5.7%
THEB	59	3.5	11	68.4%	31.6%	7.5	77.1%	22.9%
MEAN	*4,405*	*18.4*	*28,018*	*77.5%*	*22.5%*	*3,569.2*	*70.5%*	*29.5%*

Panel B: Control Stocks (Huang and Stoll, 1996)

	Market Cap	Stock	Trading Volume			Number of Trades		
	€ million	Price (€)	€ '000	% MoR	% Alternate	Count	% MoR	% Alternate
STM	3,355	4.6	15,157	100.0%	0.0%	2,635.1	100.0%	0.0%
CNAT	3,635	1.3	5,333	100.0%	0.0%	2,265.3	100.0%	0.0%
ABI	25,439	15.9	75,884	100.0%	0.0%	7,245.6	100.0%	0.0%
HRMS	10,652	101	12,193	100.0%	0.0%	1,546.2	100.0%	0.0%
OMT	84	4	15	100.0%	0.0%	9.2	100.0%	0.0%
TAM	81	6.8	29	100.0%	0.0%	23.5	100.0%	0.0%
AMG	184	6.9	2,989	100.0%	0.0%	903.5	100.0%	0.0%
SMTPC	117	20	20	100.0%	0.0%	9.4	100.0%	0.0%
DEVG	63	3.5	156	100.0%	0.0%	111.0	100.0%	0.0%
MEAN	*4,846*	*18.2*	*12,420*	*100.0%*	*0.0%*	*1,638.8*	*100.0%*	*0.0%*

Table 5. Summary Statistics

This table summarizes the trading activity, liquidity, and price efficiency variables described in Section 5.3 for the treatment stocks and the corresponding control stocks generated based on Huang and Stoll (1996). We report the mean values of the different variables for the pre-event period i.e. between 1 December 2008 and 13 January 2009, and the post-event period i.e. between 26 January 2009 and 28 February 2009.

	9 Stocks				4 Stocks			
	Treatment Group		Control Group		Treatment Group		Control Group	
	Before	After	Before	After	Before	After	Before	After
Total Volume (€ '000)	28,040	30,668	12,420	11,472	63,608	69,637	27,142	25,326
Total Trades	3545.7	3778.7	1638.8	1617.4	8013.2	8534.1	3423.0	3452.3
Quoted Spread: Inside (%)	1.153	1.097	1.302	1.568	0.140	0.198	0.272	0.191
Quoted Spread: Maximum Volume (%)	1.639	1.097	1.302	1.568	0.244	0.198	0.272	0.191
Quoted Depth: Inside (%)	12,865	12,380	15,880	10,445	23,553	23,247	31,119	18,170
Quoted Depth: Maximum Volume (%)	13,258	12,380	15,880	10,445	24,419	23,247	31,119	18,170
Effective Spread: Inside (%)	1.886	1.122	1.200	1.351	0.203	0.199	0.230	0.168
Effective Spread: Local (%)	2.350	1.122	1.200	1.351	0.233	0.199	0.230	0.168
Realized Spread 10: Inside (%)	1.623	0.820	0.827	0.906	0.082	0.047	0.086	0.048
Realized Spread 30: Inside (%)	1.601	0.784	0.785	0.849	0.061	0.019	0.056	0.018
Realized Spread 60: Inside (%)	1.540	0.744	0.797	0.828	0.048	0.008	0.036	-0.006
Realized Spread 10: Local (%)	1.935	0.820	0.827	0.906	0.079	0.047	0.086	0.048
Realized Spread 30: Local (%)	1.894	0.784	0.785	0.849	0.017	0.019	0.056	0.018
Realized Spread 60: Local (%)	1.853	0.744	0.797	0.828	-0.087	0.008	0.036	-0.006
Price Impact 10: Inside (%)	0.263	0.303	0.372	0.444	0.121	0.152	0.144	0.120
Price Impact 30: Inside (%)	0.285	0.338	0.415	0.501	0.142	0.180	0.174	0.150
Price Impact 60: Inside (%)	0.346	0.378	0.402	0.523	0.155	0.191	0.194	0.174
Price Impact 10: Local (%)	0.415	0.303	0.372	0.444	0.154	0.152	0.144	0.120
Price Impact 30: Local (%)	0.456	0.338	0.415	0.501	0.216	0.180	0.174	0.150
Price Impact 60: Local (%)	0.498	0.378	0.402	0.523	0.320	0.191	0.194	0.174
Autocorrelation 30: Inside	0.056	0.049	0.061	0.068	0.065	0.053	0.075	0.083
Autocorrelation 30: Maximum Volume	0.063	0.049	0.061	0.068	0.074	0.053	0.075	0.083
Autocorrelation 300: Inside	0.106	0.093	0.110	0.108	0.110	0.104	0.133	0.111
Autocorrelation 300: Maximum Volume	0.115	0.093	0.110	0.108	0.111	0.104	0.133	0.111
Variance Ratio 30/300: Inside	0.213	0.168	0.220	0.203	0.239	0.192	0.241	0.212
Variance Ratio 30/300: Maximum Volume	0.220	0.168	0.220	0.203	0.236	0.192	0.241	0.212

Table 6. Cross Market Arbitrage

This table reports the summary statistics on the arbitrage opportunities arising on the two Euronext order books during the pre-event period i.e., between 1 December 2008 and 13 January 2009, and their resolution. Section 6.1 describes how we identify each abitrage opportunity. Unique Instances are the average daily frequency of arbitrage opportunities on the two order books betwen 08:01 and 16:29, Negative Spread Time is the total amount of time during the trading session when the markets are crossed, Magnitude of Negative Spread is the frequency with which the negative bid-ask spread is equal to one tick, two ticks, three ticks, four ticks, and five or more ticks, and Trading Volume is the average daily volume which can be attributed towards resolution of the arbitrage opportunities.

Stock	Unique	Negative	Magnitude of Negative Spread					Trading
	Instances	Spread Time	1 Tick	2 Ticks	3 Ticks	4 Ticks	5+ Ticks	Volume
DEXI	138.1	12.8%	45.3	16.6	9.5	7.2	59.5	1,145,164
FOR	183.8	11.5%	74.5	25.2	8.9	3.1	72.1	2,361,274
ISPA	622.0	6.0%	307.8	227.2	43.7	28.4	14.9	14,185,211
UNBP	166.1	3.0%	36.7	22.2	10.2	13.0	84.1	1,968,394
GLPG	2.4	4.3%	0.0	0.1	0.0	0.0	2.2	7,558
ONCOB	0.3	1.6%	0.0	0.0	0.0	0.0	0.3	990
RCUS	0.8	9.5%	0.0	0.1	0.0	0.0	0.7	9,020
VRKP	1.2	6.7%	0.0	0.1	0.0	0.8	0.3	4,351
THEB	0.6	2.6%	0.0	0.1	0.0	0.0	0.4	1,553
MEAN	*123.9*	*6.4%*	*51.6*	*32.4*	*8.0*	*5.8*	*26.1*	*2,187,057*

Table 7. Impact on Trading Activity and Quoted Liquidity

This table reports the impact of the introduction of single order book on trading activity and quoted liquidity. The variables are defined in Section 5.3. We estimate a difference-in-difference regression for each of these variables. We report the coefficient of the variable interacting the event dummy (which equals one for all days after 13 January 2009 and zero otherwise) with the treatment dummy (which equals one for all treatment stocks and zero for all control stocks). We employ stock and day fixed effects and double cluster standard errors by stock and day. Figures in parentheses indicate t-statistics.

	9 Stocks		4 Stocks	
	Level	Log	Level	Log
Total Volume	3,637,650*	0.080	8,080,792	0.144
	[1.708]	[0.277]	[1.486]	[0.671]
Total Trades	260.6	0.153	498.9	−0.006
	[1.022]	[0.644]	[0.762]	[−0.040]
Quoted Spread: Inside	−0.323	0.090	0.139***	0.563***
	[−1.462]	[0.687]	[2.708]	[4.030]
Quoted Spread: Maximum Volume	−0.808**	−0.322**	0.035	0.023
	[−2.149]	[−2.450]	[0.811]	[0.152]
Quoted Depth: Inside	4,979	0.005	12,752**	0.386**
	[1.569]	[0.030]	[2.335]	[1.990]
Quoted Depth: Maximum Volume	4,589	0.017	11,906**	0.416**
	[1.475]	[0.105]	[2.215]	[2.252]

Notes: ***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 8. Impact on Trading Liquidity

This table reports the impact of the introduction of single order book on trading liquidity. The variables are defined in Section 5.3. We estimate a difference-in-difference regression for each of these variables. We report the coefficient of the variable interacting the event dummy (which equals one for all days after 13 January 2009 and zero otherwise) with the treatment dummy (which equals one for all treatment stocks and zero for all control stocks). We employ stock and day fixed effects and double cluster standard errors by stock and day. Figures in parentheses indicate t-statistics.

	9 Stocks		4 Stocks	
	Level	Log	Level	Log
Effective Spread: Inside	−0.926*	−0.145	0.058	0.222*
	[−1.650]	[−1.151]	[1.439]	[1.720]
Effective Spread: Local	−1.394*	−0.375**	0.028	0.073
	[−1.875]	[−2.485]	[0.671]	[0.580]
Realized Spread 10: Inside	−0.894	−0.376**	0.004	−0.174
	[−1.587]	[−2.199]	[0.132]	[−0.677]
Realized Spread 30: Inside	−0.893	−0.547***	−0.004	−0.138
	[−1.606]	[−2.853]	[−0.132]	[−0.417]
Realized Spread 60: Inside	−0.836	−0.615***	0.001	−0.365
	[−1.456]	[−3.052]	[0.045]	[−1.024]
Realized Spread 10: Local	−1.211*	−0.517**	0.006	−0.042
	[−1.657]	[−2.575]	[0.201]	[−0.136]
Realized Spread 30: Local	−1.190*	−0.643***	0.040	0.004
	[−1.662]	[−3.142]	[0.944]	[0.013]
Realized Spread 60: Local	−1.153	−0.725***	0.135	−0.299
	[−1.558]	[−3.589]	[1.036]	[−0.907]
Price Impact 10: Inside	−0.032	0.123	0.054***	0.271***
	[−0.443]	[0.879]	[2.611]	[2.998]
Price Impact 30: Inside	−0.033	0.161	0.062**	0.242**
	[−0.421]	[0.996]	[2.360]	[2.370]
Price Impact 60: Inside	−0.090	0.024	0.057**	0.215**
	[−0.897]	[0.167]	[2.201]	[2.036]
Price Impact 10: Local	−0.183	−0.164	0.022	0.030
	[−1.372]	[−0.916]	[1.289]	[0.408]
Price Impact 30: Local	−0.204*	−0.133	−0.012	0.009
	[−1.750]	[−0.743]	[−0.334]	[0.129]
Price Impact 60: Local	−0.241**	−0.214	−0.107	−0.003
	[−2.025]	[−1.431]	[−0.839]	[−0.040]

Notes: ***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 9. Impact on Price Efficiency

This table reports the impact of the introduction of single order book on price efficiency. The variables are defined in Section 5.3. We estimate a difference-in-difference regression for each of these variables. We report the coefficient of the variable interacting the event dummy (which equals one for all days after 13 January 2009 and zero otherwise) with the treatment dummy (which equals one for all treatment stocks and zero for all control stocks). We employ stock and day fixed effects and double cluster standard errors by stock and day. Figures in parentheses indicate t-statistics.

	9 Stocks		4 Stocks	
	Level	Log	Level	Log
Autocorrelation 30: Inside	−0.015	0.201	−0.020	−0.303
	[−1.576]	[0.286]	[−1.051]	[−1.557]
Autocorrelation 30: Maximum Volume	−0.021*	−0.233	−0.029	−0.282
	[−1.959]	[−0.298]	[−1.614]	[−1.609]
Autocorrelation 300: Inside	−0.011	−1.444*	0.015	−0.025
	[−0.677]	[−1.896]	[0.732]	[−0.117]
Autocorrelation 300: Maximum Volume	−0.020	−1.545*	0.015	−0.068
	[−1.259]	[−1.751]	[0.688]	[−0.329]
Variance Ratio 30/300: Inside	−0.028	−0.224	−0.019	−0.017
	[−1.250]	[−1.016]	[−0.498]	[−0.079]
Variance Ratio 30/300: Maximum volume	−0.035	−0.206	−0.016	0.063
	[−1.336]	[−0.770]	[−0.448]	[0.250]

Notes: ***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.